

Research paper

Human-in-the-loop reinforcement learning for dynamic soaring: A trajectory planning and control integrated system

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ABSTRACT

Increasing the endurance of fixed-wing unmanned aerial vehicles has long been a designer's dream. Inspired by the behavior of albatrosses, dynamic soaring holds significant potential for achieving perpetual unpowered flight. However, traditional approaches face challenges in implementing dynamic soaring in real-world scenarios. Prior studies simplified kinematics and control inputs, and separated design of the optimizer and controller. To this end, we propose a human-in-the-loop framework based on deep reinforcement learning to enable automated execution of dynamic soaring. First, pseudospectral method is utilized to generate the optimal trajectory for learning. Then, deep the reinforcement learning algorithm interacts with a high-fidelity environment, sending commands to the lower-level controller. Humans contribute domain expertise and design reward functions and evaluation criteria to assist the agent in trajectory search. Our results demonstrate the autonomous execution of dynamic soaring without human intervention, with enhanced energy collection in thin-shear wind fields. Importantly, our findings are more congruent with practical applications of real-world gliders compared to optimal control theory. The scope of this study also contains simple robustness tests which indicate the agent's resistance to wind disturbances and deviations in the initial state. This research demonstrates the feasibility of integrating automatic planning and control, accelerating the transition of dynamic soaring from simulation to real-world applications.

1. Introduction

1.1. Motivation

In recent years, the rapid advancement and frequent updates of onboard electronic devices in aviation have significantly broadened the applications of drones (Pieczyński et al., 2024). Among the numerous challenges developers face in the drone design process, one of the most critical is enhancing the drone's range and endurance. To address this pivotal challenge, energy-saving measures have been pursued, encompassing the optimization of aerodynamic shapes to enhance the lift-to-drag ratio and the exploration of alternative energy sources, such as hydrogen, as a viable substitute for conventional aviation fuel. In contrast, researchers have adopted a novel approach, focusing on the elongation of endurance by harnessing natural energy resources. Sachs and Grüter (2020).

The most common air currents commonly used for harvesting energy are shear wind and updrafts (Bencatel et al., 2013). The latter mimics birds in mountain valleys which harvest thermals, and this flight pattern has given rise to "cross-country soaring" competitions (Edwards and Silverberg, 2010). Recent research has successfully

presented and demonstrated a mature framework using hierarchical reinforcement learning to solve the core problem in ascending thermal plumes (Notter et al., 2023). In contrast to updrafts, wind shear increases with height in the form of airspeed. Its scenarios have provided the impetus for non-powered glider researchers (Mir et al., 2018a; Bousquet et al., 2017). The inspiration for this energy absorption comes from birds that soar thousands of kilometers over the ocean, including albatrosses, which are remarkable as they hardly need to flap their wings or stop to feed (Nevitt et al., 2008). Therefore, mimicking the maneuvering operations of albatrosses is deemed to be especially beneficial for a fixed-wing aircraft to stay airborne for long without consuming energy. Such periodic maneuvers have further been categorized by researchers as dynamic soaring.

Extensive research has been conducted on dynamic soaring to imitate albatrosses. As a result, a continuous flight mode has been proposed for unpowered gliders in shear winds (Sachs, 2016). This mode consists of four stages: upwind climb, high-altitude turn, downwind dive, and low-altitude turn. Similarly, theoretical possibilities for achieving permanent unpowered flight include the '8' shape and other traveling patterns proposed by Liu et al. (2017). However, it

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Nomenclature	
AR	Aspect ratio
C_D, C_L, C_Y	Coefficient of drag, lift and lateral force
C_l, C_m, C_n	Moment coefficient of roll, pitch and yaw
e	Oswald factor
$\mathbf{f}$	Vector of aerodynamic forces
h	Height
$\mathbf{g}$	Vector of gravity
$\mathbf{I}_3$	3-dimensional identity matrix
$J(\cdot)$	Objective function
$\mathbf{L}$	Lift force vector
M	Pitch moment
N	Number of interpolation points
p, q, r	Angular velocity expressed in the body coordinate system
S	Wing area
V_a	Velocity relative to the wind
$\mathbf{v}$	Velocity vector
x_N, x_E, x_D	Positions expressed in the North-East-Down coordinate system
α	Angle of attack
β	Angle of sideslip
θ	Pitch angle
γ	Flight-path angle
ρ	Density
ψ	Heading angle
$\boldsymbol{\tau}$	Vector of aerodynamic torques
τ	Dimensionless time variable
$\delta_a, \delta_e, \delta_f, \delta_r$	Deflection of aileron, elevator, flap and rudder
Subscripts	
c	Command output by the agent to the controller
i, j	Index variable of the interpolation point.
ref	Reference wind speed at the specific altitude
t	Time steps

is important to note that not all shear wind conditions allow for the completion of the Rayleigh cycle. Bousquet (2018) pointed out that there exists a minimum shear wind required for the successful execution of the dynamic soaring maneuver. More specifically, the maximum reference wind speed and the thickness of the shear layer induce changes in the tilt angle and cruising radius of the Rayleigh cycle. Drawing on the optimization findings of previous studies, the glider's maximum speed is continuously enhanced by blending air-flow with varying velocities. Sachs et al. (2021) observed that the aerodynamic characteristics of a straight wing in the high subsonic region significantly limit the attainment of maximum speed. Simulation verification suggests that a swept wing mitigates the rise of drag in compressibility regions and has the potential to enhance maximum speed performance.

1.2. Challenge

Optimization theory has significantly advanced the understanding of dynamic soaring mechanisms and provided a foundation for developing effective control strategies. Most existing work faces challenges in implementing trajectory optimization in natural environments. Considering that energy collection is highly dependent on the attitude of glider with respect to the wind, a compromise was made in our previous work by adopting an unpowered strategy referred to as “upwind climbing and downwind diving” (Wang et al., 2022). However, there are several

challenges to harvest energy. Firstly, for the sake of convergence of solvers, the motion is simplified in previous optimization work through the omission of pitch or angle of attack, with the input vector comprising the lift coefficient. Such a simplification leads to modeling errors, as the mapping from actuator deflection to angle is missing, not to mention that the lift coefficient is determined not only by the angle but also by the damping interference of the angular velocity. Secondly, treating the roll angle and lift coefficient as actuators assumes that these values can be promptly and accurately reached. In reality, as indicated by the analysis based on the optimal control solution, the controller must undergo rapid changes and persist in extreme actuator positions (Sachs and Grüter, 2021). The quickness, overshoot, and steady-state error of the trajectory tracking controller remain to be considered. Thirdly, as mentioned above, dependence on the framework of optimal control theory alone may result in unexpected trajectories if the glider deviates from the equilibrium point under wind disturbances or sensor noise. Currently, it is difficult to design a controller capable of simultaneously maintaining unpowered flight and achieving perfect tracking of the optimal trajectory in the shear wind.

Wind conditions and the operational space within the environmental range still pose a challenge for consistently imitating the maneuver of dynamic soaring. Researchers have gained empirical understandings of the dynamic soaring mechanism and harvested the energy from shear winds on the windward side of ridges (Bronz et al., 2021). From the perspective of unmanned aerial vehicles (UAVs) path planning, this involves maximizing energy harvesting and decision making (Chakrabarty and Langelaan, 2011). For robust control, it involves achieving more stable flight in unpredictable wind fields (Kim and Perez, 2023). Additionally, the time spent executing on the Rayleigh cycle is relatively limited, whether in a poor wind environment or within the compressible region of high-speed flight. Preliminary experiments validating dynamic soaring problems in the real world have been conducted in Silva and Frew (2016). Attempts by autopilots to adjust position and airspeed require the controllers to apply thrust to minimize the error between the actual and expected trajectory, which consumes power from the propulsion system. Trajectory optimization and lower-level tracking controllers are designed separately, leading to errors between the reference trajectory and the real-time position during flight tests. Stringent wind field requirements and rapid maneuvering conditions impose more precise demands on the trajectory optimization and controls of the avionics system. To the best of our knowledge, there has been few works done to integrate trajectory planning and bottom-level tracking actuators into a system for comprehensive consideration.

1.3. Novelty and contributions

To address these challenges, Fig. 1 illustrates a general scheme of the proposed human-in-the-loop (HITL) method based on deep reinforcement learning (DRL), and highlights the contribution of this study. The prior methods solve optimal control problems based on simplified dynamics models. They generate flight trajectories that satisfy the specified objectives and constraints. However, we do not want to simplify the dynamics model and controller. Instead, we aim to construct a high-fidelity environment. The environment is capable to accommodate various aircrafts and application scenarios through the adjustment of wind and aerodynamic parameters. DRL is utilized to handle trajectory planning and control tasks simultaneously. Due to the high dimensionality of the state and action spaces, HITL is introduced to involve human participation in an active learning manner. Results from optimal control theory are used to screen the optimized trajectories, design performance metrics and redefine evaluation criteria, thereby enhancing the physical significance of the trajectory. Furthermore, the incorporation of expert knowledge improves efficiency and helps prevent trapping in locally optimal trajectories.

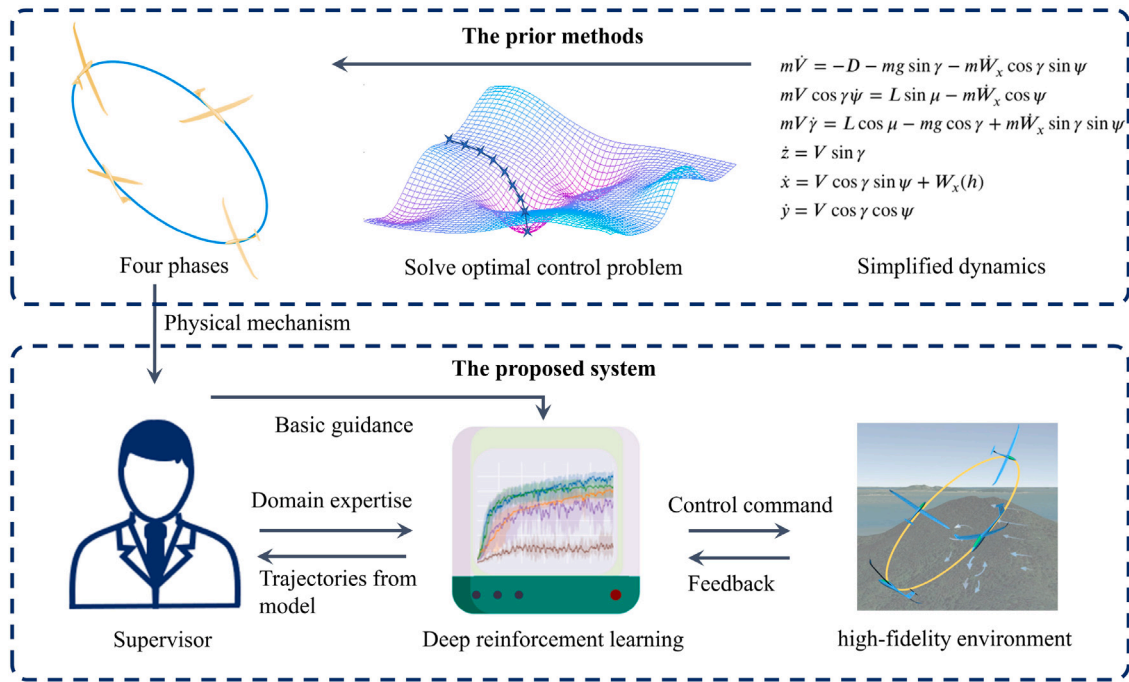


Fig. 1. General schema of the proposed human-in-the-loop method based on deep reinforcement learning and differences between the prior methods and the proposed system.

Relying on human control for prolonged periods frequently exceeds human endurance limits. Thus, our goal is to integrate the dynamic soaring into the Actor-Critic framework to develop an autonomous cruising system that replaces human operation. The trained agents can perform tasks in scenarios with random wind fields and can handle situations where the states of the aircraft deviate caused by estimation errors or disturbances from the environment.

In summary, this study makes the primary contributions as follows:

- Developed a high-fidelity simulation environment that retains the complex dynamics and controllers of gliders.
- Propose a framework to implement HITL methods based on DRL for dynamic soaring.
- Incorporate domain expertise into the RL algorithm and allow humans to participate in evaluating the quality of the agent's decisions.
- Achieve the integration of trajectory planning and control, guiding the controller to complete the trajectory.
- The agent automatically executes dynamic soaring, which requires no human intervention and exhibits a degree of disturbance resistance.

The rest of this paper is organized as follows. Previous work on dynamic soaring is presented in Section 2, which also identifies the limitations of optimization methods in implementing strategies during actual flight processes. Section 3 formulates the traditional approach to dynamic soaring and the mechanism of wind energy harvesting serves as domain expertise for the subsequent DRL work. We construct the high-fidelity environment in Section 4. Our iterative framework is introduced in Section 5, focused on the details of the HITL method and DRL algorithm. The experimental results are presented in Section 6, followed by robustness tests, indicating the feasibility of applying this approach. Finally, the contributions of this work are summarized, and future prospects are discussed in Section 7.

2. Related work

2.1. Optimal control methods

Numerous significant contributions have been made in trajectory optimization to emulate the dynamic soaring of albatrosses. Essentially,

as an optimal control problem, these studies consider nonlinear optimizers to generate optimal trajectories for dynamic soaring. The multiple shooting method was used to discretize the nonlinear differential equations (Sachs et al., 1989), and software was employed to solve for maximum altitude gain. Zhao and Qi (2004) discretized the continuous-time curve into a finite time sequence using the direct collocation method to minimize cycle time or parameters such as wind gradient slope. This approach transformed the trajectory planning problem into a large-scale nonlinear programming problem, which was subsequently solved by Sparse Nonlinear OPTimizer (SNOPT) (Gill et al., 2005). Gao and Liu (2017) reduced the dimensionality by utilizing Dubins curves, potentially compromising global optimality but significantly reducing computation time. However, these studies only computed energy-gathering trajectories for reference. Whether the glider can accurately enter the anticipative state space requires further investigation. Mir et al. (2018b) mathematically proved the nonlinear controllability characteristics along the optimal trajectory using geometric control theory, addressing this concern. Moreover, the nonlinear controllability and stability of the optimal trajectory are not mutually exclusive. The contraction theory-based technique was then used to establish the conclusion that dynamic soaring with reduced wind shear requirements is inherently unstable (Mir et al., 2021). The authors propose utilizing optimal control solutions as a reference to design controllers that generate pitch and roll commands, thereby promoting stable glider operation on the optimal trajectory.

Recently, there have been several studies focusing on UAV control in dynamic soaring or windy environments. Haghighi et al. (2024) propose an efficient computational framework, which applies pseudospectral method to accurately solve optimal trajectory strategies in a windy environment. A Type-2 Fuzzy Logic Controller is introduced combined with PID for UAVs with wind disturbances (Moali et al., 2024). Similarly, A Gaussian Pseudospectral method and a cascade dual loop scheme are adopted, achieving robust control and effective trajectory generation (Pan et al., 2022). Additionally, a robust MPC scheme in unknown scalar fields is proposed, where reference trajectories are generated by an RL algorithm (Li et al., 2024). These works exhibit outstanding control effectiveness, even when modeling errors arise from simplifications and disparities between assumed and

actual behavior exist. By combining trajectory optimization with other controllers, researchers aim to enhance the UAV's ability to harness wind energy effectively.

2.2. Reinforcement learning

Significant progress has been made in utilizing reinforcement learning (RL) to harness energy in wind fields. Montella and Spletzer (2014) obtained a novel RL controller by imitating the sampling results from a nonlinear optimizer, resulting in improved performance compared to the baseline teacher controller. Impressive demonstrations by Reddy et al. (2016, 2018) showcased the ability of energy-constrained gliders to learn how to exploit and explore energy solely through their interaction with unknown thermal updraft environments.

Nevertheless, reinforcement learning is not universally capable of handling all tasks without conditions. Due to the challenges faced by Q-learning in dealing with continuous glider states, deep neural networks, which excel in approximating arbitrary nonlinear functions, are employed to estimate the Q-value, an evaluation metric that guides the aircrafts' action in specific states (Novati et al., 2019). The concept of updating the action strategy at each step based on the Q-value has been categorized by researchers as a value-based learning method. Similar to state processing, the optimal actions can be composed of roll angles and pitch angles, which can be discretized into multiple action groups (Montella and Spletzer, 2014). In practice, discretizing the state and action spaces is a compromise to address the physical problems. A more optimal approach is to adopt the sample mean estimation (Schulman et al., 2015), which avoids the exponential mapping from the continuous space of glider angle inputs to the discrete space and reduces the computational cost of fitting the neural network. The approach of representing the input actions through a probability distribution function and subsequently optimizing the parameterized policy using numerical algorithms is known as a policy-based reinforcement learning method (Notter et al., 2023). In fact, the Actor-Critic framework has been proposed to encompass the benefits of each approach, handling the limitations of discrete actions while being able to update Q-values and policies in a single step (Wang et al., 2016b).

Recent advancements in DRL have significantly improved the autonomy and adaptability of UAVs and other robotic systems in complex environments. Traditional model-based controllers face challenges when dealing with nonlinear dynamics, environmental uncertainties, and real-time adaptation requirements. To address these limitations, Ramezani et al. (2023) introduce an LSTM-based Model Predictive Control (MPC) framework integrated with the Deep Deterministic Policy Gradient (DDPG) algorithm for UAV path planning in complex environments. This approach enhances convergence speed while reducing failure rates. Similarly, Baek et al. (2022) propose a hybrid control strategy that combines linear quadratic regulator (LQR) controllers with ensemble DRL methods. By leveraging prior knowledge, this method improves sample efficiency and robustness. Meanwhile, SAC algorithms have become a key method for continuous control. Variants such as the Soft Actor Critic Gaussian Mixture Model (SAC-GMM) enable skill refinement in trajectory space (Nematollahi et al., 2022), while SAC achieves reactive collision avoidance in dynamic scenarios (Nakhleh et al., 2023). Additionally, Garg incorporated vision-based DRL for UAV path planning and focuses on assessing the generalization from simulation to real-world deployment (Garg et al., 2024). These approaches demonstrate performance in real-world deployments. Such innovations highlight DRL's potential to overcome traditional bottlenecks in UAV control.

2.3. Human-in-the-loop-methods

Artificial intelligence strives to integrate the advantages of human intelligence, aiming for effective collaboration between humans

and machines (Mosqueira-Rey et al., 2023). Integrating expert insights into algorithms creates a HITL-augmented approach, enhancing the reliability and rationale of outputs (Sadeghianasl et al., 2024). This methodology has found application in diverse fields, including medical diagnosis, visual recognition, classification, and recommendation systems (Wang et al., 2016a; Liu et al., 2023). Active human feedback has been employed to refine the efficacy of semi-automatic annotation (Bonet-Jover et al., 2023). Liu et al. (2019) demonstrated the value of pre-selecting datasets for fine-tuning algorithm parameters, which expedites convergence. Expert-machine interaction reduces search spaces, simplifying the process of obtaining clean training data, making it more efficient (Holzinger, 2016).

To the best of the authors' knowledge, no existing studies have employed DRL with HITL in dynamic soaring. The potential for the automated execution of dynamic soaring is worthy of thorough exploration. We believe that this work can reawaken interest in the achievement of dynamic soaring in the real world.

3. Trajectory optimization based on pseudospectral method

This section presents the classical method of generating Rayleigh cycle using pseudo-spectral methods. The theoretical basis obtained will be utilized as domain expertise, which guides the DRL algorithm to enable the search for energy-harvesting trajectories. The optimal trajectories generated by optimal control principles will be further compared in detail with the trajectories generated iteratively based on the proposed framework in Section 5.

3.1. Mathematical model of dynamic soaring

The point mass model is used to describe flight trajectories. Given the representation of shear wind as W_x , the following partial differential equations can be derived for flights in a wind field:

$$m\dot{V} = -D - mg \sin \gamma - m\dot{W}_x \cos \gamma \sin \psi \quad (1)$$

$$mV \cos \gamma \dot{\psi} = L \sin \mu - m\dot{W}_x \cos \psi \quad (2)$$

$$mV\dot{\gamma} = L \cos \mu - mg \cos \gamma + m\dot{W}_x \sin \gamma \sin \psi \quad (3)$$

$$\dot{z} = V \sin \gamma \quad (4)$$

$$\dot{x} = V \cos \gamma \sin \psi + W_x(h) \quad (5)$$

$$\dot{y} = V \cos \gamma \cos \psi \quad (6)$$

Where V , γ , and ψ denote airspeed, flight path angle and yaw, respectively. In order to control the nonlinear motions, control variables need to be introduced. Angular velocity was employed in Mir et al. (2018b) for controllability analysis to adjust the pitch and roll of the glider. Generally, for the purpose of simplification, the coefficient of lift (C_L) and roll angle are commonly chosen in trajectory optimization methods, although Control command at a lower level offers a more realistic reproduction of the physical processes in experimental flight. In this context, the introduction of u completes the form of the nonlinear dynamics system as $\dot{x} = \hat{f}(x, u)$.

$$L = -\frac{1}{2}\rho V^2 S C_L \quad (7)$$

$$D = -\frac{1}{2}\rho V^2 S C_D \quad (8)$$

Where C_D is the drag coefficient. In Deittert et al. (2009), the drag coefficient is the quadratic relation of the lift coefficient and is inversely proportional to the maximum lift-drag ratio. In our experiment, we determined the drag coefficient as the inverse proportional function of aspect ratio, which was formulated as follows:

$$C_D = C_{D0} + \frac{(C_L - C_{L0})^2}{\pi e A R} \quad (9)$$

Where C_{L0} represents the lift coefficient at zero angle of attack, while C_{D0} is the minimum of the lift-drag curve. In fact, both approaches illustrate the reduction of lift-induced drag with an increase in aspect ratio, essentially conveying the same concept.

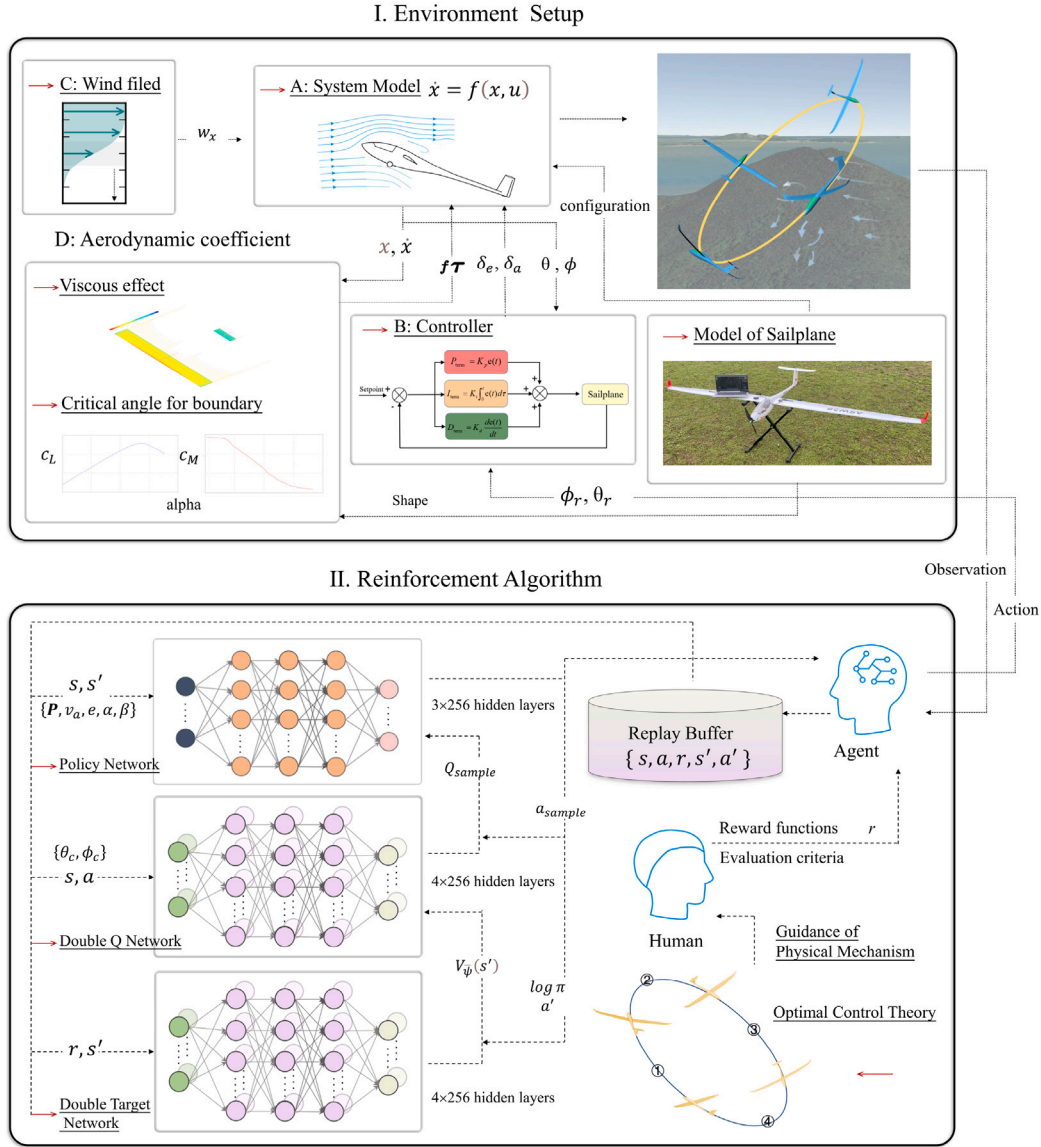


Fig. 2. The iterative framework which details two subsystems. Subsystem I represents the established environmental subsystem, whereas Subsystem II represents the combination of reinforcement learning and human involvement. The environment subsystem comprises various sub-modules, including the wind field module, dynamics system module, controller module, and reverse engineering module. Dashed lines indicate data transmission between different modules and subsystems. In II, the left side depicts the structure of DRL, while the bottom right corner represents the knowledge acquired by humans from the optimal trajectory, which is subsequently transferred to DRL through the HITL method as expert domain knowledge.

3.2. Formula for optimal control problem

Determining the optimal objective is not restricted to a single criterion and depends on the purpose of our investigation. In order to find the critical conditions for an extended-duration cruise, the minimum reference wind speed was computed by [Bonnin et al. \(2015\)](#). [Silva and Frew \(2016\)](#) explored the minimum thrust necessary to conserve fuel during flight. [Gao and Liu \(2017\)](#) prioritized maximizing airspeed gain as a criterion to maximize energy collection. In this paper, to obtain the altitude gain as much as possible under poor wind conditions is of great significance. The formulation is presented as follows:

$$J = \min [-h_{t_f}] \quad (10)$$

$$\mathbf{x}(t)_{t_f} = \mathbf{x}(t)_{t_0} \quad (11)$$

Eq. (10) represents the objective function of dynamic soaring. It indicates that the goal is to minimize the negative of the final height h_{t_f} , meaning the optimization aims to maximize the final altitude at the

terminal time. When the wind is sufficiently strong, the endpoint of the flight will be higher than the starting point. Conversely, when the wind is not strong enough to complete the entire Rayleigh cycle, the endpoint of the flight will be lower than the starting point. While t_f and t_0 represent the starting and ending time of the optimization, respectively. For the case of finding the maximum height gain, we maintain other constraints but remove the restriction on the height. For state and input variables, the following path constraints are given:

$$V_{\min} < V < V_{\max}, \quad \psi_{\min} < \psi < \psi_{\max} \quad (12)$$

$$\gamma_{\min} < \gamma < \gamma_{\max}, \quad x_{\min} < x < x_{\max} \quad (13)$$

$$y_{\min} < y < y_{\max}, \quad \phi_{\min} < \phi < \phi_{\max}, \quad z \geq 0 \quad (14)$$

where ϕ denotes the roll and z is the altitude of the aircraft. The inequalities impose various limits on the variables over the trajectory, such as prohibiting attitude angles and airspeed from exceeding specified ranges, with the aim of keeping the flight vehicle in a stable flight condition. In our article, the spectral method is applied with

high-order global basis functions over the entire computational domain to discretize the state and control variables, transforming the optimal control problem of solving $x = f(x, u)$ into a nonlinear programming problem (NLP). The equations involve solving for the state variable x and control input u over time, denoted as $x(\tau)$ and $u(\tau)$, where τ is a normalized time variable.

$$x(\tau) \approx X(\tau) = \sum_{i=0}^N L_i(\tau) X(\tau_i) \quad (15)$$

$$u(\tau) \approx U(\tau) = \sum_{k=1}^N L_k(\tau) U(\tau_k) \quad (16)$$

Where lagrange polynomials are used to approximate the state variables and control inputs. This approximation helps transform the continuous problem into a discrete one. The polynomials are mathematically described as:

$$L_i(\tau) = \prod_{j=0, j \neq i}^N \frac{\tau - \tau_j}{\tau_i - \tau_j}, \quad i = 1, 2, \dots, N \quad (17)$$

Each polynomial $L_i(\tau)$ is associated with a specific time node τ_i and they satisfy the property in Eq. (17). This allows us to approximate the values of $x(\tau)$ and $u(\tau)$ smoothly between discrete points. To ensure higher accuracy, discrete points τ_i in Eq. (15) and (16) no longer follow the equidistant distribution. Instead, the roots of the linear combination of high-order Legendre polynomials, $P_{n-1}(\tau) + P_n(\tau)$, are chosen as collocation points. In this research, the optimal control software GPOPS-II (Patterson and Rao, 2014) and the nonlinear programming solver (SNOPT) are employed to solve the optimal control problem for dynamic soaring.

4. Construction of the high-fidelity environment

In this section, a detailed introduction to the full-scale environment that interacts with DRL, encompassing the wind field, dynamics and control system, is provided. Similar to an automated control system, the agent is characterized as a parameterized guider, while the glider acts as an actuator to execute commands. The environment is comprised of everything outside the agent. The agent is a machine to be educated, and its final decisions are intensely influenced by its interactions with the environment. Consequently, the effectiveness of applying the agent to actual flight is largely determined by the realism of the environment.

Thus, the high-fidelity environment is constructed, as depicted in Fig. 2I. The glider is firstly reverse-engineered and parameterized. Slight modifications are made to the size to satisfy the requirements of dynamic soaring. Then, the dynamic system with the controller, is modeled based on the measurement. This system receives information from the wind field module and provides the necessary state information of the glider for DRL. In part D, XFlr and Athena Vortex Lattice (AVL) are adopted to estimate viscous-related parameters and aerodynamic coefficients. The static derivatives and rotation-rate-dependent coefficients are calculated using the vortex lattice method and the embedded software Xfoil. State information is collected by this module, and feedback on the six-dimensional forces and moments is offered. An exponential wind field is utilized to generate the required wind for dynamic soaring, with the option of adding the von Kármán wind turbulence in part C. A PID controller with tuned parameters is created in part B to control the glider's motion.

It is noted that compared to the real-world environment, the simulation environment depicted in Fig. 2I still has minor limitations. In Part C, the wind field is approximated by superimposing an empirical model with a stochastic process. However, in real-world scenarios, wind fields are typically generated by complex interactions, such as the collision of incoming flow with hills or the friction of airflow over the sea surface. While numerical simulation results can help minimize the gap between simulated and real wind fields (Lu et al., 2025), the computational speed of Computational Fluid Dynamics (CFD) remains relatively slow.

On the other hand, in Part D, vortex lattice theory is used to calculate aerodynamic coefficients to accelerate the solution process. The numerical discrepancies between the calculated aerodynamic forces and real-world values can be attributed to the inherent limitations of the vortex lattice method.

4.1. Flight dynamic system model

The state information of a rigid body moving in space is described by its position and attitude. Here the orientation of the vehicle with respect to the inertial system is expressed in quaternions.

$$q = \begin{bmatrix} q_w \\ q_v \end{bmatrix} = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} \quad (18)$$

where q_w denotes the real part of the quaternion, while $q_v = [q_x \ q_y \ q_z]^T$ represents the imaginary part. The dynamic system using vector notation is presented to make the mathematical equations more condensed:

$$\begin{bmatrix} mI_3 & 0 \\ 0 & J \end{bmatrix} \begin{bmatrix} {}_B \dot{v} \\ {}_B \dot{\omega} \end{bmatrix} = \begin{bmatrix} -m[\omega]_{\times} & 0 \\ 0 & [J\omega]_{\times} \end{bmatrix} \begin{bmatrix} {}_B v \\ {}_B \omega \end{bmatrix} + \begin{bmatrix} m_B g \\ 0 \end{bmatrix} + \begin{bmatrix} {}_B A f \\ {}_B \Delta \tau \end{bmatrix} + \begin{bmatrix} {}_B f \\ {}_B \tau \end{bmatrix} \quad (19)$$

where J represents the inertia matrix. The symbol B denotes the projection of physical quantities in the body frame and ω is the vector representation of the Euler angles. Unlike the simplified set of differential equations in Section 3.1, the lateral force is now exerted on the glider. Furthermore, in Eq. (19), the solution for the Euler angles is no longer skillfully circumvented as in Eq. (1)~(6), which bypasses the rotational motion equations. Most importantly, the complex calculations of aerodynamic forces and torques are determined by the following formulas, rather than directly as input quantities simply disregarding the positive and negative gains brought by angles and angular velocities.

$$\begin{aligned} C_L &= C_{L0} + C_L^a \cdot \alpha + C_L^{\delta_f} \cdot \delta_f + C_L^{\delta_e} \cdot \delta_e + \frac{c}{2V_a} (C_L^{\dot{\alpha}} \cdot \dot{\alpha} + C_L^q \cdot q) + C_L^M \cdot M \\ C_Y &= C_Y^{\beta} \cdot \beta + C_Y^{\delta_a} \cdot \delta_a + C_Y^{\delta_r} \cdot \delta_r + \frac{b}{2V_a} (C_Y^p \cdot p + C_Y^r \cdot r) \\ C_l &= C_l^{\beta} \cdot \beta + C_l^{\delta_a} \cdot \delta_a + C_l^{\delta_r} \cdot \delta_r + \frac{b}{2V_a} (C_l^p \cdot p + C_l^r \cdot r) \\ C_D &= C_{D0} + \frac{(C_L - C_{L0})^2}{\pi e A R} + C_D^{\delta_f} \cdot \delta_f + C_D^{\delta_e} \cdot \delta_e + C_D^{\delta_a} \cdot \delta_a + C_D^{\delta_r} \cdot \delta_r + C_L^M \cdot M \\ C_m &= C_{m0} + C_m^a \cdot \alpha + C_m^{\delta_f} \cdot \delta_f + C_m^{\delta_e} \cdot \delta_e + \frac{c}{2V_a} (C_m^{\dot{\alpha}} \cdot \dot{\alpha} + C_m^q \cdot q) + C_m^M \cdot M \\ C_n &= C_n^{\beta} \cdot \beta + C_n^{\delta_a} \cdot \delta_a + C_n^{\delta_r} \cdot \delta_r + \frac{b}{2V_a} (C_n^p \cdot p + C_n^r \cdot r) \end{aligned} \quad (20)$$

The formulas assume that aerodynamic forces are generated due to positive and negative changes in control inputs or angles, such as actuator deflections and angle of attack. The torques are also influenced by the control surface deflections and flight conditions. In addition, the forces and moments take into account the effects of angular velocities p, q and r . The lateral coefficients are retained. These factors are typically neglected in traditional trajectory optimization.

4.2. Flight control

The aircraft's motion in the wind field operates without a control loop, making it an open-loop system. The PID controller, renowned for its maturity, efficiency and ease of deployment in real aircraft, is also extensively used in various machine devices beyond aviation.

As our work progresses towards aircraft applications, we adopt a PID controller as the underlying controller to serve the upper-level guidance system:

$$P(t) = K_P \left[e(t) + \frac{1}{T_I} \int e(t) dt + T_D \frac{de(t)}{dt} \right] \quad (21)$$

Where K_P , T_I , and T_D are the proportional, integral, and derivative coefficients respectively, which play crucial roles in determining the tracking speed of the reference signal, steady-state error, and response lag. The tracking performance of the controller is of utmost importance for the upper-level guidance system. In this experiment, we evaluate the performance of the PID controller using a $\pm 10^\circ$ step signal.

DRL does not integrate the environment into the optimization process directly. The distinct advantage allows us to tailor the complexity and flexibility of environment modeling. This aspect is evident in our ability to introduce more challenging conditions, including random wind fields, sensor noise, and other factors. These uncertain conditions are hard to be formulated through a series of differential equations in optimal control methods. Unlike the flight dynamics previously presented, our interactive environment avoids simplifying assumptions in regard to the wind, motion, and controller.

5. Proposed HITL method based on deep reinforcement learning

5.1. The HITL iterative framework

Dynamic soaring can be tricky when relying on humans or AI systems alone. Thus, an iterative framework is proposed detailed in Fig. 2 which combines humans, DRL algorithm and environment integrating wind field, dynamics and control system. The algorithm starts with arbitrary initialized states which should be aligned with the attitude obtained from the optimal control theory. The reward function is designed to maximize energy gain and promote energy accumulation and retention throughout dynamic soaring. Simultaneously, it is essential to guide the glider through turns in the high altitude region. Simultaneously, it is essential to direct the glider to execute turns within the high altitude region. In reality, during the initial stages, the algorithm often converges towards an acceptable yet suboptimal trajectory. Towards the trajectory's end, the state frequently reaches the maximum or minimum angle of attack or roll angle, potentially resulting in stalling or rolling. This outcome arises from the changes of the angle of attack during the upwind or tailwind phase, which aids in energy collection. Furthermore, swift turns during the downwind descent augment energy acquisition and curtail energy consumption during the turning process.

In the subsequent step, we introduce the concept of the HITL approach as shown in the red dotted box of Fig. 3. Through comparing the state approaching the termination time, the trajectory that closely aligns with the pseudospectral method is manually selected as the initial state for the subsequent iteration. Based on the four phases of dynamic soaring, reward functions and assessment criteria are carefully designed. These interventions, guided by human expertise, integrate model training and testing outcomes into the algorithm, creating new enhanced intelligence through human participation. This process of training and iteration is disclosed in Algorithm 1, which continues until the algorithm converges and the agent successfully fulfills the Rayleigh criterion task.

5.2. Stochastic strategy

In reinforcement learning, there is no need to explicitly establish an accurate mathematical model of the system dynamics because it is an indirect policy optimization method. In the initial stage, it uses a policy to collect a series of state(s)-action(a) pairs and rewards (R). Then it maximizes the cumulative return at current and future discounted

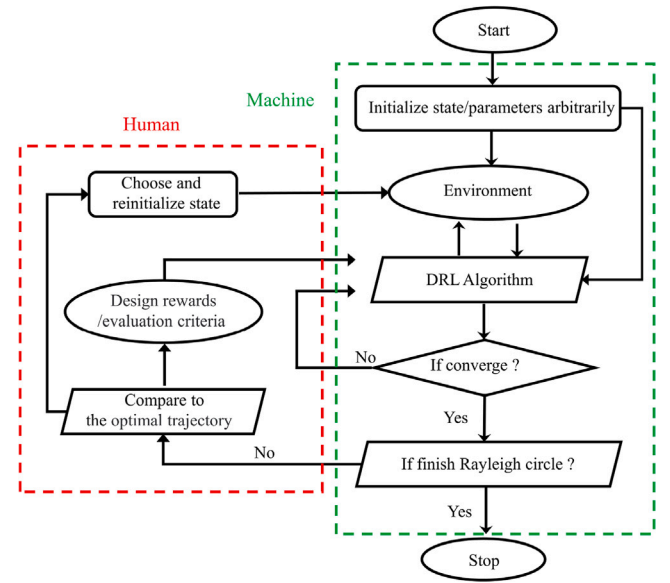


Fig. 3. The flow chart of the proposed iterative framework where domain expertise is embedded through interaction with the DRL algorithm.

Algorithm 1: The proposed HITL method.

Input: The environment and configuration of a glider

```

1 if not finish dynamic soaring then
2   while not converge do
3     | Execute Algorithm 2
4   end
5   Human: design  $\{r_E, r_M, r_P, r_T\}$ 
6           setting evaluation criteria
7           reinitialize  $\{P, v_a, e, \alpha, \beta\}$ 
8 end
Output: Agent

```

moments, based on a specially designed reward function. Reinforcement learning iteratively optimizes a new strategy by evaluating the next observed state (s') and R , improving the return. This strategy can provide the optimal action based on the current state. Among these concepts, the Markov process serves as a general framework for describing the interaction between the agent and the environment, defined by a tuple $(S, \mathcal{A}, P(\cdot), R(\cdot), \gamma)$. Here, $\mathcal{A}$ is the set of action spaces, representing actions that the agent takes to change its state. The state transition probability ($P(\cdot)$) describes the probability of reaching the next state (s') after executing an action in the current state. The state transition probability distribution of the motion system is independent of the previous state trajectory and depends only on the current ($s-a$).

In cases of poor exploration and generalization, maximum entropy is incorporated into the reward to address this limitation. The entropy regularization term in entropy-augmented rewards is utilized to incentivize the agent to adopt diverse strategies. Additionally, in noisy environments, agents trained with entropy-augmented functions demonstrate a certain level of robustness against interference. The coefficient α is employed to achieve a balance between the weights assigned to the two terms:

$$r(s, a) = R(s, a) + \alpha \mathcal{H}(\pi(\cdot | s)) \quad (22)$$

Where $\mathcal{H}(\pi(\cdot | s))$ is introduced as entropy regularization at state s , and α is a hyperparameter that controls the trade-off between exploration and exploitation. Higher entropy encourages more exploration. Eq. (22)

leads to a stochastic strategy, which is then transformed to maximize the entropy objective function:

$$J(\pi) = E_{\pi} \left[\sum_{t=0}^T \gamma^t r(t+1) \right] \quad (23)$$

$$= E_{(s_t, a_t) \sim \rho^{\pi}} \sum_{t=0}^T [r(s_t, a_t) + \alpha H(\pi(\cdot | s_t))] \quad (24)$$

$$= \sum_{t=0}^T E_{(s_t, a_t) \sim \rho^{\pi}} [r(s_t, a_t) + \alpha H(\pi(\cdot | s_t))] \quad (25)$$

$J(\pi)$ represents the expected return obtained by following the policy. ρ^{π} is the distribution of the policy. The discount factor γ allows for adjusting the weight given to immediate and future rewards. In extreme cases where γ approaches 1, our agents adopt a long-term perspective, disregarding the risks associated with immediate actions. Such agents are more courageous, exhibiting increased exploration of future scenarios, as they can tolerate negative rewards from current actions and vice versa.

In fact, using a stochastic policy is necessary for the dynamic soaring problem. Ensuring the generality and convergence of the algorithm is essential for efficiently deploying our algorithm framework. However, these two requirements are often contradictory. For instance, certain spatial information crucial for the agent to understand the glider may not be used to characterize the dynamics due to the variations in flight regions during soaring.

For MDPs, there exists a deterministic greedy optimal policy that satisfies the Bellman optimality equation (BOE) and maximizes the state value. The deterministic policy exhibits outstanding performance due to the accurate characterization of dynamics (Barth-Maron et al., 2018). If state information perceived by agents is insufficient to describe the dynamics, the environmental information is only partially observable and the agent may encounter state aliasing. Due to the absence of the Markov property, the optimal policy ceases to be deterministic. In such a situation, a stochastic policy outperforms a deterministic policy as it alleviates the challenges of convergence when facing state aliasing and local optima.

5.3. Maximize entropy soft policy iteration

Soft policy iteration is proposed as a general algorithm for maximizing entropy reward. The optimal policy is obtained by alternating between policy evaluation and improvement. Soft policy evaluation guarantees the convergence of the true Soft Q-function (Sutton and Barto, 1998) defined as follows:

$$Q_{soft}(s, a) = r_t + \gamma E_{s_{t+1} \sim p_{s+1}} [V_{soft}(s_{t+1})] \quad (26)$$

$$V_{soft}(s) = E_{a_t \sim \pi(s_t)} [Q_{soft}(s_t, a_t)] + H(\pi(s_t)) \quad (27)$$

$$= E_{a_t \sim \pi(s_t)} [Q_{soft}(s_t, a_t) - \log \pi(s_t)] \quad (28)$$

The soft Q-function, Q_{soft} , combines the immediate reward and the future return, discounted by γ . The state value V is the average of these return values taking into account future scenarios. The returns can vary among different trajectories. The entropy term $H(\pi(s_t))$ ensures that the policy π remains stochastic. This is particularly useful to try out different actions to find better strategies.

To exploit the characteristics of the distribution of Q-function, we define the action-value function as the negative energy function $\mathcal{E} = -\frac{1}{\alpha} Q(s, a)$. Minimizing the Kullback–Leibler (KL) divergence between the distribution and the previous energy-based policy distribution generates a new policy.

$$\pi_{new} = \arg \min_{\pi' \in \Pi} D_{KL} \left(\pi'(\cdot | s_t) \parallel \frac{\exp(Q^{\pi_{old}}(s_t, \cdot))}{Z^{\pi_{old}}(s_t)} \right) \quad (29)$$

Where $Z^{\pi_{old}}$ is used for normalizing the distribution. The policy is further improved based on the probability distribution derived from the updated value function. Through iterative processes of formulas Eqs. (28) and (29), it converges mathematically to $\pi_{optimal}$. This policy maximizes the entropy-based state value, promoting more balanced probabilities among the output actions and enhancing the exploratory nature of the algorithm. When faced with a large number of state–action spaces in the environment of dynamic soaring that are difficult to collect, the presence of entropy disperses the action probability distribution, accelerating action search and enhancing algorithmic convergence.

Enhancing agent exploration is critical in dynamic soaring. Optimal control results demonstrate that the Rayleigh ring requires action reversal, characterized by alternating actions in opposite directions. The exploratory capability of stochastic policies with entropy objectives has been highlighted (Haarnoja et al., 2018b). While incorporating a Gaussian distribution into DDPG (Lillicrap et al., 2015) also encourages random exploration, this approach does not fully address the problem of multiple peaks in the Q function. When the reward function encompasses more than a goal, it indicates a multimodal task. The benefit of the stochastic strategy is that the output is determined according to multiple extreme values of the probability distribution. Therefore, these advantages of maximizing entropy soft policy iteration make it applicable to training agents to perform dynamic soaring.

5.4. Dynamic soaring in soft A-C algorithm

When applying RL methods to high-fidelity dynamic soaring environments, we need to consider not only continuous states but also continuous actions. The policy iteration method in Section 5.3 has been mathematically proven to converge for solving the discrete maximum entropy optimal policy (Haarnoja et al., 2018a). However, the actions were discretized and value-based RL was used (Reddy et al., 2016). Adopting this scheme is impractical, as the Q-function for state–action pairs has excessively high dimensionality. In the context of the dynamic soaring studied in this paper, function approximators are employed to extend the solution from discrete problems to continuous ones. Similar to value-based RL, deep neural networks (DNNs) are utilized to approximate the V and soft Q in Eqs. (27) and (28), addressing the storage and computational challenges associated with high-dimensional discrete data:

$$J_V(\psi) = \mathbb{E}_{s_t \sim D} \left[\frac{1}{2} (V_{\psi}(s_t) - \mathbb{E}_{a_t \sim \pi_{\phi}} [Q_{\theta}(s_t, a_t) - \log \pi_{\phi}(a_t | s_t)])^2 \right] \quad (30)$$

$$J_Q(\theta) = \mathbb{E}_{(s_t, a_t) \sim D} \left[\frac{1}{2} (Q_{\theta}(s_t, a_t) - \hat{Q}(s_t, a_t))^2 \right] \quad (31)$$

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim p} [V_{\psi}(s_{t+1})] \quad (32)$$

Eq. (32) describes the method for calculating the target Q-value, $\hat{Q}(s_t, a_t)$. It consists of two parts: the immediate reward $r(s_t, a_t)$ and the discounted expected value of the next state $V_{\psi}(s_{t+1})$. Then we minimize the difference between the current $Q_{\theta}(s_t, a_t)$ and the target $\hat{Q}(s_t, a_t)$ as shown in Eq. (31). The Q-function is used to compute the expected return under the current policy to train the value network $V_{\psi}(s_t)$. The experience replay buffer, denoted as D , stores memories of interactions with the environment. The soft Bellman residual is minimized in the form of the squared error. Stochastic gradient descent is employed to optimize both Eq. (30) and (31). A target value network $V_{\bar{\psi}}$ is fixed during training and updated every time step.

In policy-based reinforcement learning, the optimal policy is iteratively enhanced by parameterizing it to maximize the expected cumulative reward. In fact, the A-C framework introduced in this paper combines value-based and policy-based iteration to solve for the optimal policy. To be embedded in high-fidelity dynamic soaring environments, the SAC algorithm is prioritized, which approximates the energy-based model (EBM) policy distribution:

Table 1
Coefficients of the reward function designed for different phases.

Coefficients	Phase 1	Phase 2	Phase 3	Phase4
c_1	0.16	0.2	2.5	3.9
c_2	-5.3	-16.8	-19.5	26.0
c_3	-1.6	-2.6	-10.6	-16.4
c_4	0.0	-3.8	0.0	153.0
c_5	0.0	7.30	15.8	0.0
c_6	-10.75	12.70	23.0	-25.4
c_7	0.0	-30.4	-104.0	-217.0
c_8	31.0	38.3	57	94
c_9	968.0	90.3	172.0	312.0

$$J_{\pi}(\phi) = \mathbb{E}_{s_t \sim D} \left[D_{KL} \left(\pi_{\phi}(\cdot | s_t) \parallel \frac{\exp(Q_{\theta}(s_t, \cdot))}{Z_{\theta}(s_t)} \right) \right] \quad (33)$$

$$= \mathbb{E}_{s_t \sim D} \left[\mathbb{E}_{\mathbf{a}_t \sim \pi_{\phi}} [\alpha \log(\pi_{\phi}(\mathbf{a}_t | s_t)) - Q_{\theta}(s_t, \mathbf{a}_t)] \right]$$

Where ϕ is the parameter set of the policy network. $J_{\pi}(\phi)$ is used to optimize the policy such that it maximizes the expected return while maintaining a certain level of entropy. Eq. (33) uses the Kullback-Leibler (D_{KL}) divergence to measure the difference between the current policy $\pi_{\phi}(\cdot | s_t)$ and a target distribution $\frac{\exp(Q_{\theta}(s_t, \cdot))}{Z_{\theta}(s_t)}$ derived from the Q-function, where $Z_{\theta}(s_t)$ is a normalization factor.

Optimizing these equations is equivalent to maximizing the state value function. However, directly sampling $\mathbf{a}_t$ can lead to difficulties in gradient computation. To address this, we employ DNNs to update policy parameters and enable independent action sampling from noise, leveraging their gradient accessibility. By applying the reparameterization to compute the expectation with respect to noise, the complications are alleviated arising from the distribution dependence on policy parameters. The pseudocode of the soft A-C(SAC) for dynamic soaring is provided in Algorithm 2.

Algorithm 2: The SAC algorithm.

Input: $\{r_E, r_M, r_P, r_T\}$, evaluation criteria
1 Reinitialize state of the glider, $\{\psi, \tilde{\psi}, \theta, \phi\}$
2 **while not converge do**
3 **for each environment step do**
4 $\{\theta_c, \phi_c\}_t \sim \pi_{\phi}(\mathbf{a}_t | s_t)$
5 $\{P, v_a, e, \alpha, \beta\}_{t+1} \sim p(s_{t+1} | s_t, \mathbf{a}_t)$
6 $D \leftarrow D \cup \{(s_t, \mathbf{a}_t, r(s_t, \mathbf{a}_t), s_{t+1})\}$
7 **end**
8 **for each gradient step do**
9 $\psi \leftarrow \psi - \lambda_V \hat{\nabla}_{\psi} J_V(\psi)$
10 $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i)$ for $i \in \{1, 2\}$
11 $\phi \leftarrow \phi - \lambda_{\pi} \hat{\nabla}_{\phi} J_{\pi}(\phi)$
12 $\tilde{\psi} \leftarrow \tau \psi + (1 - \tau) \tilde{\psi}$
13 **end**
14 **end**
Output: θ_1, θ_2, ϕ

One crucial factor that enables the broad application of model-free RL across diverse tasks and scenarios is the sensitivity of algorithms to hyperparameters. In certain cases, with similar hyperparameters, the resulting policy can be locally optimal or even divergent. To mitigate this issue, D3QN is proposed with fewer hyperparameters (Wang et al., 2016c), which expands its applicability to a broader range of scenarios. In the context of such challenges, the SAC is adopted, which focuses on only a few parameters and enhances algorithm stability.

It should be noted that algorithms capable of continuous action have emerged rapidly, such as PPO (Schulman et al., 2017), but they also come with the inherent drawback of low sample utilization in on-policy RL. Since there is no need to take the complex environment of dynamic soaring into account, the sample is highly sensitive for model-free algorithms. Therefore, off-policy algorithms are introduced to establish an experience replay buffer for reuse. SAC combines the

characteristics of off-policy and random policy, making it particularly well-suited for aircraft applications. Unlike ground-based robots, the high cost associated with aircraft machinery leaves no room for failure or crashing. These events result in significant economic and time costs that must be avoided. Even in simulated flight, on-policy is avoided as much as possible because it requires experiences far beyond what humans need by several orders of magnitude. Thus, improving the efficiency of the sampling process is of utmost importance.

5.5. Reward design

When the task requirements are straightforward, the physical state and action spaces can be extensively explored without exceptions, and a simple reward function is adequate to derive effective strategies iteratively. In the case of complex tasks, designing a reward function is challenging as it often necessitates a profound understanding of the application context. The iterative process will quickly terminate without appropriate reward guidance. Our task is to explore the value function space as quickly as possible. However, in reality, the environment contains an immense number of state-action pairs, making it unfeasible to collect them all. A suitable reward function leads the agent into the correct state-action-reward chain, reducing the necessity to explore excessive state-action spaces and expediting the convergence of the algorithm. Therefore, in our article, the reward function is thoughtfully constructed, which comprises the energy gain term r_E , penalty term r_P , mechanism-guiding term r_M , and terminal term r_T .

$$r = r_E + r_M + r_P + r_T \quad (34)$$

The energy gain term involves the aspiration of the glider to maximize energy collection or minimize energy consumption in increased or decreased energy phases.

$$r_E = c_1 \Delta E \quad (35)$$

In the pseudospectral method, the principles that climbing during the upwind phase and diving during the downwind phase are essential for energy collection are recognized. Based on the principle of belly-to-wind, we divide the state space into five parts: upwind climb, high-altitude turn, downwind dive, low-altitude turn, and the area outside these four parts. The dynamic soaring task significantly depends on a physical background. Therefore, within each region, the agent is guided by the principles of the Rayleigh cycle.

$$r_M = c_2 \Delta \theta + c_3 \Delta \phi + c_4 \Delta x_N + c_5 \Delta x_E + c_6 \Delta x_D \quad (36)$$

Where, Δ represents the change of states between the current moment and the previous moment, for example, $\Delta \theta = \theta_t - \theta_{t-1}$, and similarly for $\Delta \phi, \Delta x_N, \Delta x_E, \Delta x_D$. Note that these coefficients appear contradictory to the penalty term shown in the next paragraph or may conflict at different phases. For instance, persisting with a downward attitude in the downwind phase can frequently exceed the state boundary. In such a situation, short-term rewards might be negative, but accumulated rewards can be positive. Which action to take should be determined by the overall return rather than immediate rewards to avoid shortsighted decisions.

The penalty terms consist of state anomalies and exceeding the boundaries of the four stages. The action penalty for state anomalies is a variation of the constraint in the optimal control theory, serving as a warning to the agent to avoid approaching the limits of safe flight:

$$r_P = \begin{cases} p_1 = 0 & \text{if in Phase 1} \sim 4 \\ p_1 = -c_7 & \text{else} \end{cases} + \begin{cases} p_2 = 0 & \text{if in normal flight} \\ p_2 = -c_8 & \text{in danger} \end{cases} \quad (37)$$

If the glider fails to meet the state criteria, it is deemed to be in an unsafe state, i.e. $10 < h < 40$, $v_a > 3$, $|\theta| < 50$ or $|\phi| < 80$. Besides, following physics principles, the agent likely possesses

explicit objectives at destinations. The terminal term r_T for each stage is determined as follows:

$$r_T = c_9 - \begin{cases} \|\bar{P} - \bar{P}_{ini}\| + |\theta_c - \theta| + |\phi_c - \phi| & \text{if at terminal} \\ 0 & \text{else} \end{cases} \quad (38)$$

Coefficients of the reward function are listed in Table 1. In the headwind climb phase, we encourage the glider to climb, collect energy, and perform appropriate turns to transition into phase 2. The turning c_2 and c_3 are intended to maintain the aircraft's headwind climb and a slight roll attitude. c_4 and c_5 are temporarily inactive. c_6 is set to a negative value because, at this stage, we do not want the UAV to turn too sharply.

During the high-altitude turn, the glider is expected to change its direction while consuming an appropriate amount of energy. At this point, coefficients related to position become influential. c_4 and c_5 are activated to guide the aircraft to move eastward and towards the ground, indirectly completing the turn. Upon reaching this stage, we must reward the algorithm to signal that reaching phase 2 is crucial, so c_7 begins to take a non-zero value.

As the aircraft enters phase 3, it has completed the turn and is preparing for the descent phase. The main objective is to maximize wind energy for consumption in the low-altitude stage. We amplify the coefficient of the energy component, so c_1 increases by an order of magnitude, while c_6 is doubled. Due to the previous maneuvering operations, the glider is prone to crashing into the sea or experiencing excessive roll angles. We set c_8 to constrain these dangerous maneuvers.

During the low-altitude turn, c_2 and c_4 become the primary components controlling the growth of the reward function. They significantly outweigh other coefficients related to attitude and position. This is because UAV is required to climb steeply while altering its flight direction, making a sharp turn from south to north.

It is important to note that the absolute values of c_7 and c_8 continuously increase. This encourages the algorithm to prioritize long-term rewards over immediate gains, as future rewards become appealing. Guided by this principle, the absolute values of all coefficients from phase 1 to phase 4 increase, ensuring the reward function exhibits a consistent growth trend. In addition, C_9 is the largest in phase 1. This is to guide the aircraft back to the starting point, ensuring that the Rayleigh cycle forms a complete cycle.

5.6. Evaluation criteria

Although we have partitioned the reward function into four stages, the algorithm lacks knowledge of the glider's state. Therefore, we must analyze the results from optimal trajectories and manually establish evaluation criteria based on position and attitude. These criteria fall into two categories: range-based criteria, which inform the algorithm about the current stage, and endpoint-based criteria, which determine if the glider has reached the departure or entry points of phases.

Through the setting of evaluation criteria, the motion information is divided into five parts, which correspond to the four phases of dynamic soaring and the crash case. We believe that subdividing the state spaces significantly reduces the search space for the algorithm. Useless samples no longer need to be mindlessly collected. The agent collects state-action pairs within a limited information range, which leads to fast convergence to the optimal value and policy.

5.7. Parameter configuration

Within the framework of this paper, S represents the characteristic information of the glider with respect to the wind field and inertial frame. The state is of paramount importance, which should not only capture the characteristics but also be easily measurable in real-world environments. The horizontal position $P \in \mathbb{R}^2$, airspeed v_a , Euler angles

$e \in \mathbb{R}^3$, angle of attack, and sideslip angle are selected:

$$S \in \{P, v_a, e, \alpha, \beta\} \quad (39)$$

Including α and β in S is motivated by gliders encountering variable wind. In conjunction with the airspeed, these two angles indirectly capture the forces, moments, and attitudes relative to the wind.

The output of RL built upon the autopilot is an instruction that indirectly controls gliders, as the agent illustrated in the top right corner of Fig. 2II. Hence:

$$\mathcal{A} \in \{\theta_c, \phi_c\} \quad (40)$$

Where, θ_c and ϕ_c are the pitch angle and roll angle command output by the agent to the controller, respectively. Overestimation of the Q-function is a common issue in neural network-based approaches. To mitigate estimation errors, two Q-networks and two target networks are trained independently, as illustrated by the shaded portion of DNNs in Fig. 2 II. An exponentially moving average is adopted to update the target network weights with a smoothing coefficient τ . It is crucial to carefully consider the architecture and parameters of the networks as they are related to the stability of training. Including redundant features and sparse states leads to gradient explosion and vanishing in Q-function approximation. In our framework, the resulting artificial neural network for estimating the Q network has 4 hidden layers, with 256 neurons in every layer. The policy network features 3 hidden layers with 768 neurons in total, mapping state to the expectation and variance of each action.

6. Experimental results and analysis

In this section, we first present the Rayleigh cycle generated by the pseudo-spectral method and briefly analyze the necessary attitude conditions for energy harvesting in dynamic soaring. Then, a comparative experiment is conducted using HITL-based DRL versus non-HITL DRL to demonstrate that the agent struggles to guide the aircraft to complete dynamic soaring without human in the loop. The trajectories generated by the proposed method are illustrated and further compared with the optimal trajectories generated by the pseudo-spectral method. Finally, robustness tests related to random winds and simulated sensor deviations are performed.

6.1. Results from trajectory optimization

The flight region in Fig. 4 is divided into left and right parts, distinguishing the Rayleigh cycle flights in strong-wind and poor-wind conditions. Their trajectories are represented in black and red respectively. Triangular points in the figure represent the starting position, while circular points indicate the end of the flight. When the reference speed reaches 12 m/s, the endpoints in the left half lie above the starting point, indicating that the glider retains excess energy after completing one cycle of flight. This surplus energy can serve as a reserve for flying to other areas or for continuing subsequent rounds of Rayleigh cycle flight. Conversely, the right half shows energy depletion under 10 m/s conditions. To fulfill the requirements of Eq. (11), the glider must consume additional energy. The distinctive characteristic of soaring in the figure is flying below the starting point.

In fact, the complete closed-loop flight is explicitly considered and the objective function is minimized as follows:

$$J = \min [V_{w,ref}] \quad (41)$$

Adding the constraint $z_0 = z_{t_f}$, we obtain the minimum reference wind speed required to close the Rayleigh cycle as 10.78 m/s. This result further validates the energy conservation analysis at wind speeds of 10 m/s and 12 m/s shown in Fig. 4.

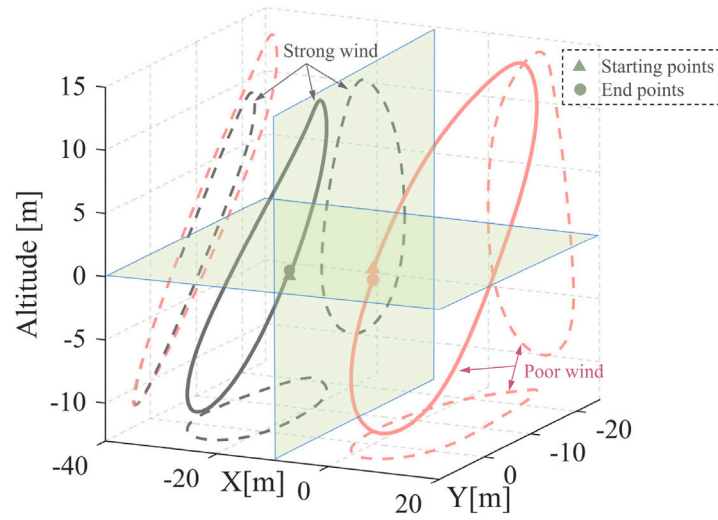


Fig. 4. Two flight perspectives with characteristics of the Rayleigh cycle. The triangles represent the starting points while the circles indicate the end points. The wind speed is stronger on the left with 12 m/s and is poor with 10 m/s on the right.

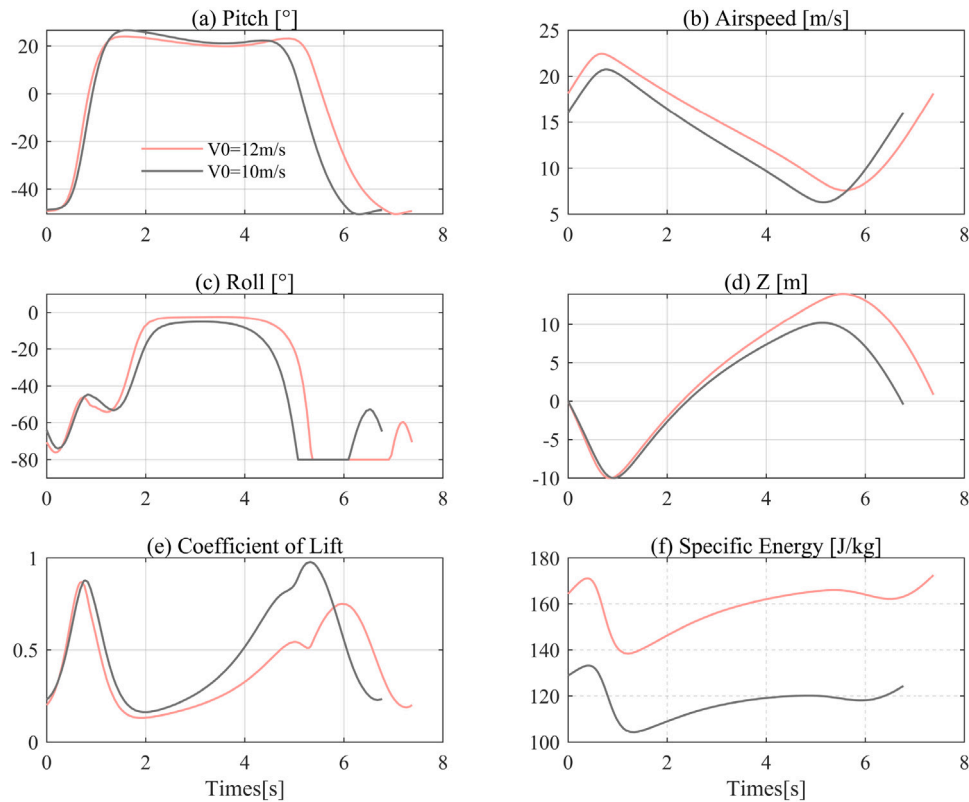


Fig. 5. The state and specific energy of the glider under different wind conditions. The red line represents strong wind conditions, while the black line denotes poor wind conditions. In d, the glider lacks sufficient energy to return to the starting point shown in the black line.

Flight state results are shown in Fig. 5. As expected, the specific energy at the endpoint is lower than at the starting point to satisfy the equality of initial and final states for the poor wind case. Conversely, for the 12 m/s case, the specific energy at the endpoint is higher. The pitch angle experiences rapid increases and decreases, corresponding to the low-altitude and high-altitude turning phases. Owing to the higher wind speed, the airspeed at the starting and ending moments is also greater than that at 10 m/s. A more comprehensive understanding of the state and trajectory for optimal dynamic soaring can be found in our previous work (Wang et al., 2022).

The optimal control inputs are shown in Fig. 5c and Fig. 5e. As discussed in the introduction, the roll undergoes rapid and drastic changes to meet the requirements of high-altitude and low-altitude turning. However, few attempts were made to design controllers to achieve accurate and fast performance. Another weakness is that, during the turning process, the lift coefficient and roll angle typically approach the aircraft's aerodynamic and state limits, posing challenges to structural load and safety margins. Furthermore, these optimizations prioritize simplified assumptions of control inputs. For instance, only the roll angle remains in lateral motion. Deflections of the control face and

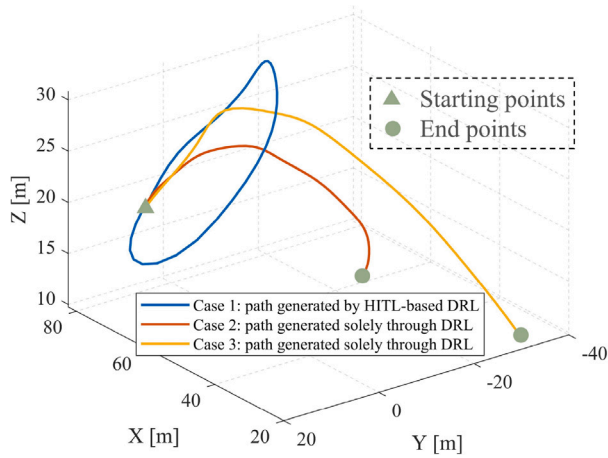


Fig. 6. Three-dimensional trajectory generated through HITL-based DRL versus non-HITL DRL. The triangle indicates the starting point of the flight, while the circle denotes the endpoint.

aerodynamic damping are not sufficiently considered leading to the loss of aerodynamic force.

Perspective view with characteristics of the Rayleigh cycle is divided into two altitude layers: one corresponds to the area above the middle layer, characterized by high wind speeds, while the other pertains to the region below the middle layer, exhibiting lower wind speeds. This characteristic flight aligns with the phenomenon mentioned by Bousquet (2018), where different momenta are mixed within the medium. More specifically, the Rayleigh cycle comprises four characteristic flight phases: upwind climb, high-altitude turn, downwind descent, and low-altitude turn. In the respective flight phases, gliders exhibit distinct attitudes that enable them to harness wind energy and minimize energy consumption. These flight attitudes and physical mechanisms are mastered by humans. Through the HITL approach, humans actively engage with the algorithm to refine the training process.

6.2. comparison experiments of HITL versus non-HITL approaches

Fig. 6 illustrates the flight trajectories guided by the agent after training with DRL and HITL-based DRL, respectively. The results of the HITL-based DRL training serve as the baseline for comparison. It is clear that generating Rayleigh cycle applying DRL alone, without expert knowledge, is a formidable challenge. Case 2 represents one of the results of training using non-HITL DRL, indicated by the orange line. Without human-in-the-loop, the agent is unable to guide the UAV to harness wind energy. In the first stage of flight, the aircraft is directed to descend too early without reaching a sufficient altitude. Furthermore, because of the lack of expert knowledge, the agent fails to lead the aircraft back to the starting point. The blue curve in the figure represents the results of HITL-DRL. It shows that with human evaluation of the agent's behavior, the UAV climbs to an enough altitude and then consumes energy. The agent chooses to give up immediate rewards to direct the glider to conduct a 180-degree turn. Then it continues to guide the aircraft to dive to harvest energy, and finally brings the aircraft back near the starting point.

We attempt to initialize an alternative attitude, as shown in Case 3, indicated by the yellow line in Fig. 7. After climbing to a certain height, the glider begins to descend while not making a complete U-turn. The agent believes that diving to gain airspeed and increase energy is the best option. These actions lead to airspeed exceeding the set boundaries and a dangerously low altitude, which ultimately results in a crash as represented by the yellow curve in Fig. 6. If there is no timely

intervention from a human operator, the agent would not direct the aircraft to climb up.

6.3. Performance of HITL-based SAC algorithm

Fig. 8a shows the trajectory of the Rayleigh cycle generated by the proposed HITL method. In the top view, the trajectory remains circular, while in the side view and the front view, it appears as two irregular ellipses. These results resemble those obtained through the pseudospectral method. The energy color map reveals that the glider predominantly gathers energy during the downwind gliding phase. The energy acquired during the upwind climb and high-altitude turns is counterbalanced by energy consumption, leading to a nearly constant total energy level. The energy curves in Fig. 9f provide additional verification of this observation. The relatively minor energy variation during the upwind climb phase can be attributed to the arbitrary initial state of the glider, with energy being utilized to adjust its attitude. The airspeed and altitude curves indicate the glider's near-identical state at the task's beginning and end. Therefore, the starting and ending points depicted in Fig. 8a almost coincide. Analysis of the roll and yaw angles still allows for the division of the glider's actions and states into four parts, aligning with the four phases in Rayleigh cycle.

The states during the flight process are illustrated in Fig. 9. The yellow line represents the actions generated by the policy, guiding the aircraft within the environment and producing the blue signals. The pitch angle curve in Fig. 9a demonstrates the ability of the controller to promptly and accurately track the commands from the agent. While this may not be prominently evident roll in Fig. 9c, as previously mentioned, reinforcement learning algorithms exhibit adaptability to imperfect environments. The agent keenly detects the errors of the roll controller and adjusts accordingly in response to the tracking error, ensuring the return to the starting point.

Additionally, we observed the algorithm's tolerance towards deviations in the initial state. It is evident in Fig. 9a, where the initial pitch angle decreases instead of increasing directly. This is because a deviation exists between the initialized and the optimal state, making a direct climb inadequate in approaching the optimal trajectory. When the agent adheres to the suboptimal strategy of direct climbing, resulting in reaching the boundaries of safe flight or crashing due to limited remaining energy, penalties are incurred. The agent, burdened with the maximum entropy task, opts to daringly explore alternative actions until it approaches the optimal trajectory rather than sticking to direct climbing, even though the exact form of the optimal trajectory in the high-fidelity environment encompassing all sub-modules is unknown to us. The agent progressively steers the glider towards the optimal trajectory, thereby ensuring the reproducibility of the Rayleigh cycle. While this is achieved in the pseudospectral method by imposing constraints on the initial and final states.

The comparison is conducted in Fig. 10 between the trajectories generated by the HITL method and obtained from the pseudospectral method. The DRL-generated trajectory exhibits smooth state transitions indicated by red arrows. In terms of area and perimeter, the optimal trajectory appears smaller in the frontal and top views, and in the three-dimensional plot, it is encompassed by the DRL-generated trajectory. The fundamental reason for this is that DRL operates in a high-fidelity environment, running on a complete aircraft model with controllers. In contrast, the pseudospectral method does not consider these aspects, as the roll angle and lift serve as input actions u that directly achieve the specified state without error. As a result, the climbs and descents in the side view appear steeper. In Fig. 10, the side view shows an inclination of 41.7° between the connecting line and the ground, whereas the pseudospectral method exhibits an angle of 64.5° as indicated by black arrows. The pseudospectral method is expected to exhibit more drastic action changes, which can be verified in Fig. 11. The glider completes the Rayleigh cycle in 6.4 s, while the system with the control module takes approximately 8 s. We also noticed that the extreme actions

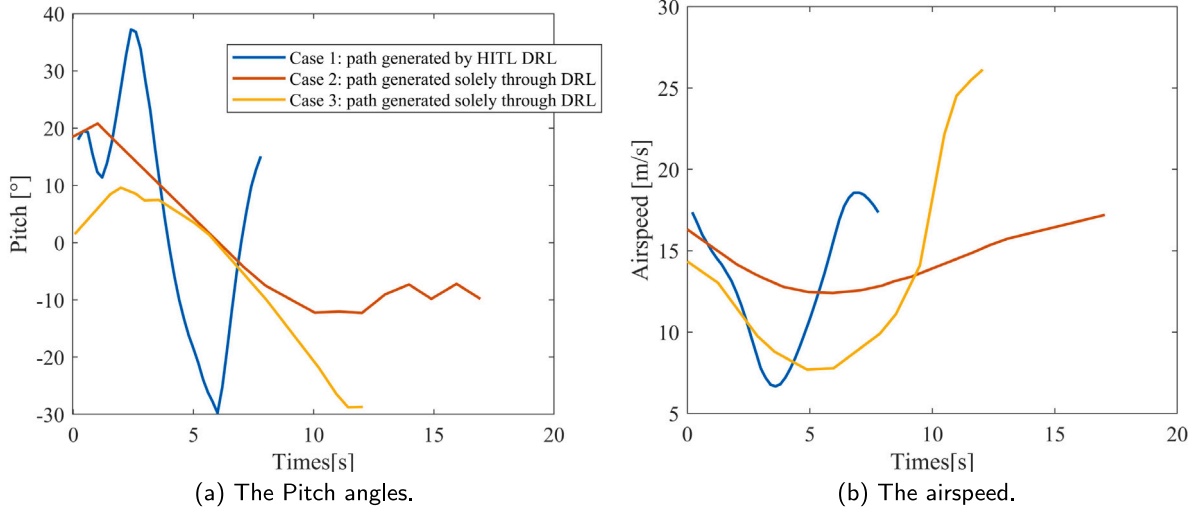


Fig. 7. Partial flight states generated through HITL-based DRL versus non-HITL DRL.

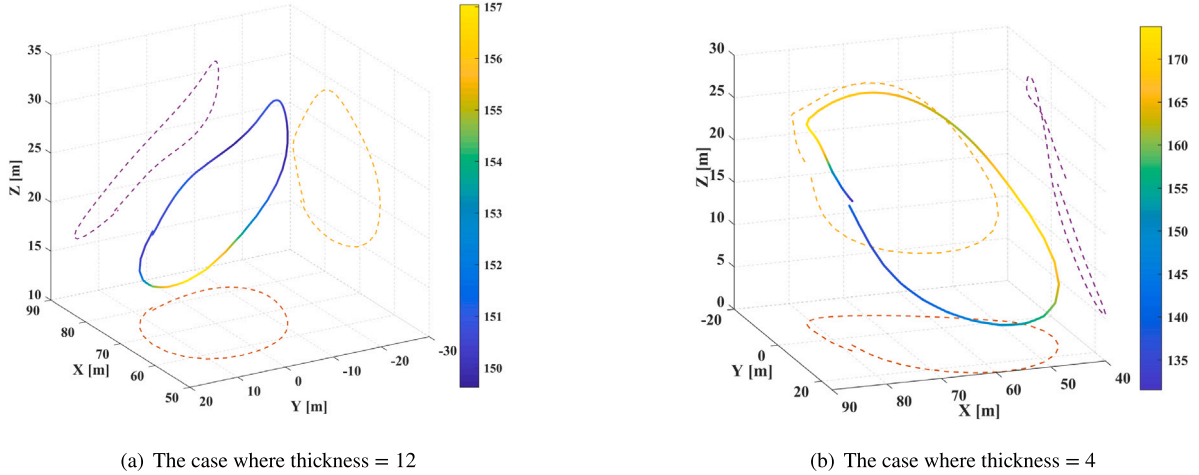


Fig. 8. The three-dimensional trajectory obtained from the HITL method. The dashed lines represent the projection on the three coordinate planes.

in the pseudospectral method are noticeably larger than those in the DRL approach. Although this is necessary for rapidly completing the Rayleigh cycle, it also places the glider in a region of high-risk flight.

The framework used for training the agent is generic. We modify the thickness of the shear layer to make the wind field more severe. By reducing the thickness from 12 to 4, we transform the wind field from linear to nonlinear. In the challenging environment, the agent decisions continued to steer the glider along the Rayleigh cycle trajectory, as depicted in Fig. 8b. In a linear wind field, the climbing altitude range is about 13 m, while in a nonlinear wind field, the climbing altitude range increases to 15 m. Due to the extreme environment, the glider took more time to complete the Rayleigh cycle.

We initialize similar state in both wind fields, but the non-linear wind field inevitably leads to differences in the trajectory. The glider again is successfully guided back onto the correct trajectory. As the thickness of the shear layer decreases, the net energy gain of the UAV after completing dynamic soaring increases, which is consistent with the findings in Bousquet (2018). In fact, the applicability of Algorithm 1 to diverse wind field types is crucial as the nature of the wind is beyond human control. For the deployment of the SAC algorithm on dissimilar

terrains, it is necessary to determine wind parameters manually, and selecting a pre-trained agent is essential.

6.4. Robustness tests

Before integrating reinforcement learning algorithms into onboard computers, it is essential to conduct a series of robustness tests. Applying modern control theory for stability tests or controllability analysis becomes challenging when dealing with a multi-module integrated environment and strategies. Hence, it is more practical to conduct disturbance-resistant tests of the strategies to ensure the completion of the Rayleigh trajectory, despite the uncertainty. The performance of the well-trained intelligent agent is evaluated under scenarios encompassing random wind fields. For situations involving angle and position deviations caused by operations, we apply bias to the initialization state.

Random wind introduces uncertainty that may cause deviations from the desired control performance and may lead to significant accidents in gliders. Therefore, the test to evaluate the robustness of the trained strategy in the wind fields is conducted. The von Karman

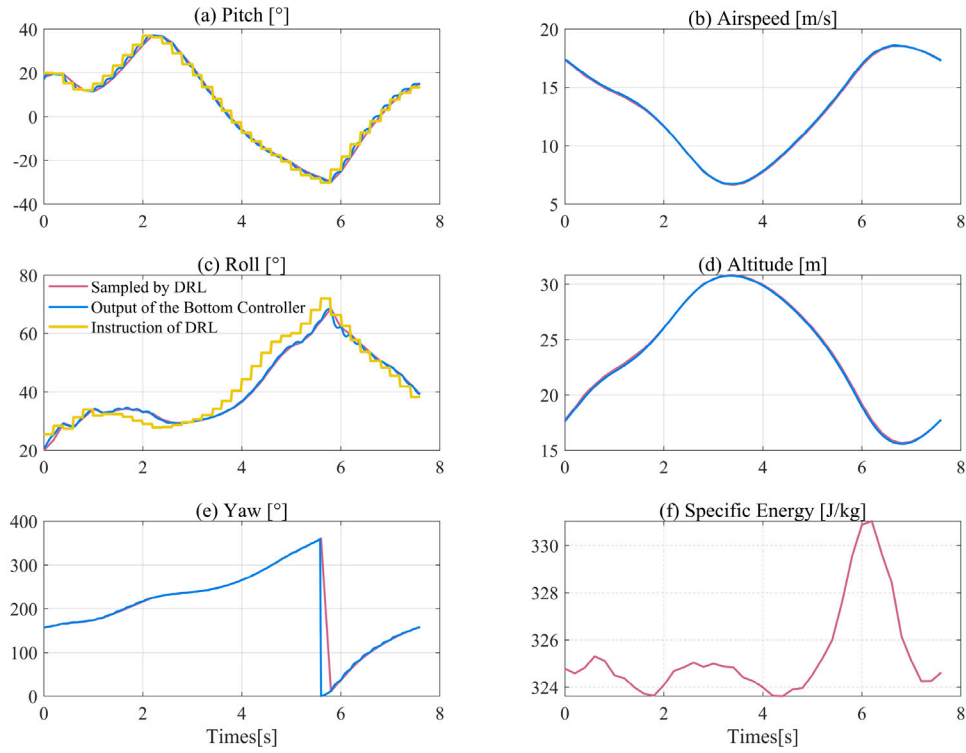


Fig. 9. The data transmitted between the agent and the environment and the state of the glider.

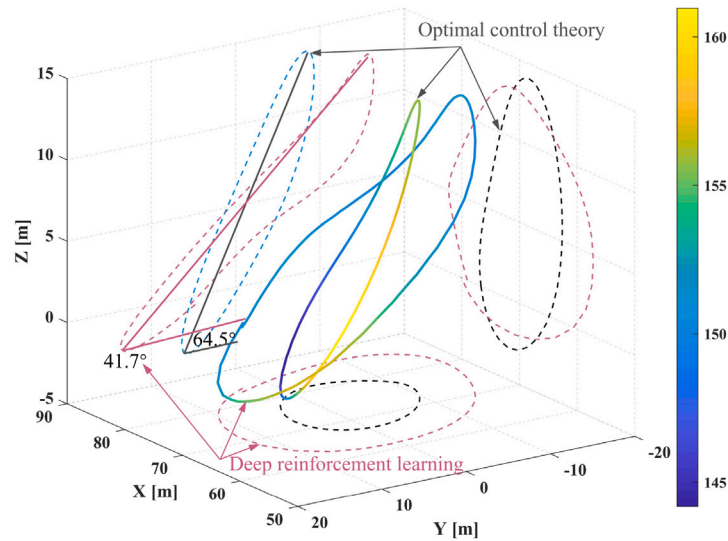


Fig. 10. The comparison between trajectories obtained by DRL and optimal control theory. The solid black line, drawn manually, represents the degree of inclination between the trajectory and the ground.

wind is adopted to introduce stochastic characteristics, as depicted in the three-dimensional wind speed in Fig. 12b. The black and green curves represent the lateral and vertical wind speeds, respectively. The vertical wind speed hovers around zero, while the lateral wind speed gradually increases up to 1 m/s. Fig. 12a and Fig. 12c display the action commands generated by the agent for PID controllers, represented by the yellow lines. At approximately 2s and 5s, the roll angle controller experiences slight deviations from the agent's actions due to the influence of increased lateral wind. However, it still exhibits a consistent tracking trend. Fig. 12e presents the side view, revealing that the perturbation caused by random wind results in a less precise

overlap between the starting and ending points compared to Fig. 8a. Nevertheless, Fig. 12d demonstrates that the altitude and airspeed still exhibit an overlap between the initial and final states, with only a slight positional deviation. The energy plot illustrates that the total energy of the complete Rayleigh cycle is almost preserved, with a slight decrease of 2.95 J/kg compared to the initial state, which matches our calculation of 2.68 J/kg for the vertical wind in the random wind field. It is evident that the decrease in energy is due to the negative work done by the vertical wind field on the glider.

The takeoff position will vary with the terrain, leading to deviations. Thus, we deliberately introduce a 5% bias in the initial position in the

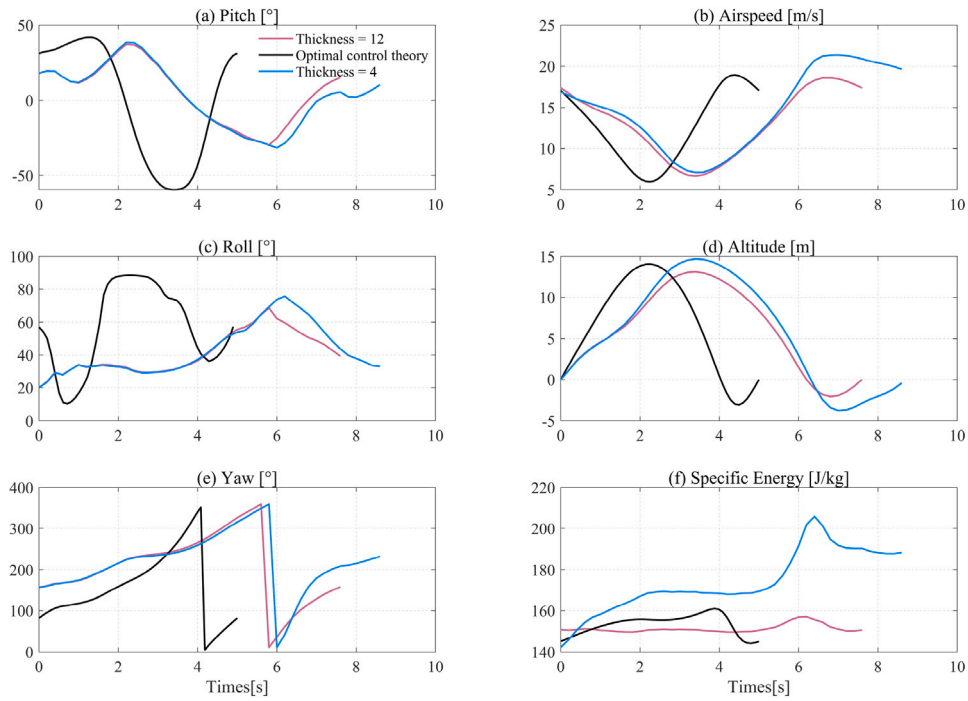


Fig. 11. The state of the glider generated through various methods and wind conditions.

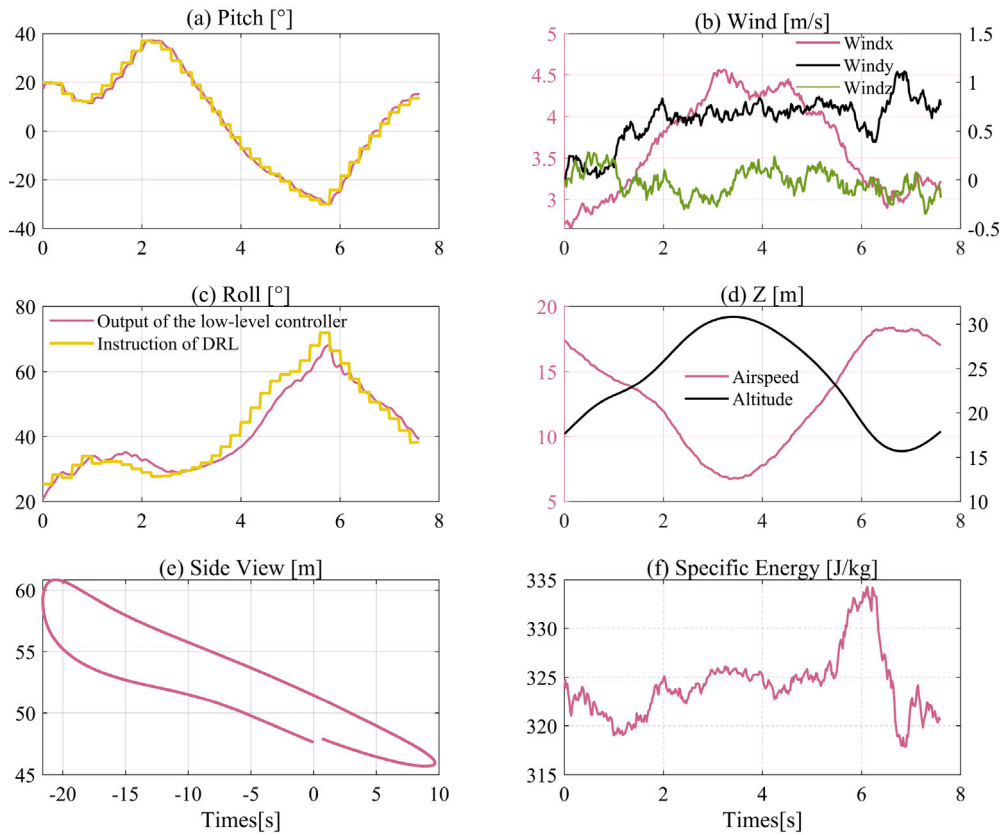


Fig. 12. The state of the glider and output of the low-level controller in the presence of random wind.

respective direction to perturb the strategy. The standard trajectory is shown as a solid blue line for comparison in Fig. 13a. Our findings indicate that deviations to the north have minimal impact on the strategy. Conversely, deviations to the east and south result in the

endpoint deviating eastward from the starting point. The simulation results remind us that positional errors in the southeast direction may cause a non-overlapping endpoint, highlighting the importance of minimizing such errors in real-world scenarios. The heading angles

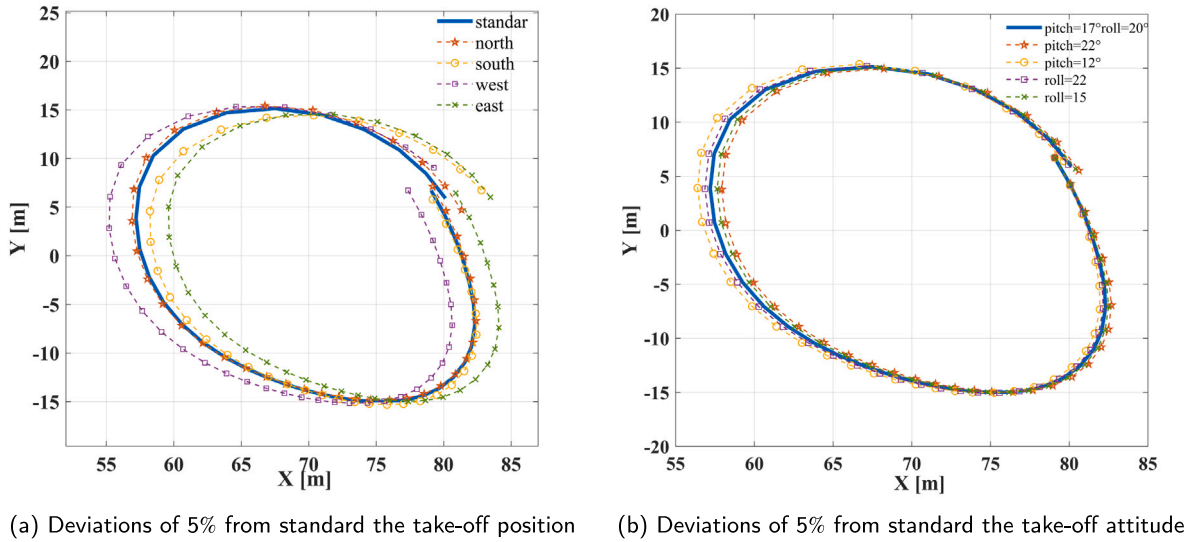


Fig. 13. Top view of trajectories with initially set bias generated by DRL.

remain in close proximity to the initial values, as depicted in Fig. 13a. The agent can complete the Rayleigh trajectory while conserving energy plot in Fig. 14a. As previously mentioned, strategies derived from DRL with maximum entropy rewards demonstrate a certain level of generalization ability, capturing multiple patterns that approximate optimal behavior.

Achieving a precise attitude corresponding to the optimal trajectory at the takeoff position is challenging. Consequently, considering the angle deviation when the glider enters the initial trajectory is needed. Modifying the initial roll and pitch angles of the glider, we evaluate the intelligent agent's ability to complete the dynamic soaring. Fig. 13b demonstrates the remarkable adaptability of the DRL method. In comparison to the reference trajectory, attitude changes do not impede the intelligent agent's ability to make correct action decisions. The angle variations are detailed in Fig. 14b, showing initial deviation from the standard takeoff angles followed by rapid convergence to the desired trajectory. The agent displays a greater tolerance for angle deviations when compared to positional deviations. Of utmost significance, it accomplishes the task with almost no energy consumption.

6.5. Limitation and discussion

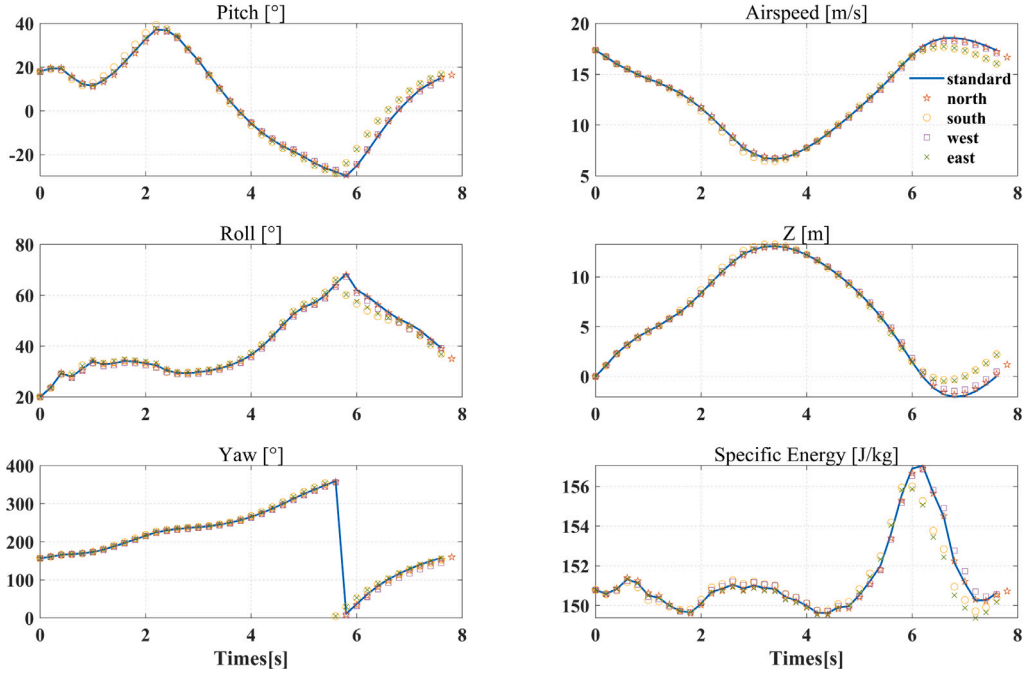
These results provide support for the deployment of DRL on the onboard computer. However, due to the inherent intricacy involved in deciphering the decision-making process of reinforcement learning, proving the mathematical convergence of the iterative framework and controllers poses a formidable challenge. A constraint of the proposed algorithm is its limited transferability. An agent trained under specific wind conditions may not be readily adaptable to arbitrary wind fields. The current scope of our work is confined to examining a single Rayleigh cycle, leaving unanswered questions regarding the agent's ability to sustain the glider in flight for extended durations.

While a real-world flight experiment represents the ultimate test of our approach, experiments are strictly restricted due to extremely demanding flight scenarios. The wind fields required for dynamic soaring are more commonly found in specific sea surfaces. Another potential site for such flights is the leeward side of mountains. This scenario also requires a ridge with a certain slope and winds over an extended period. In the future, we will find a suitable site for dynamic soaring to conduct deployment experiments of the algorithm.

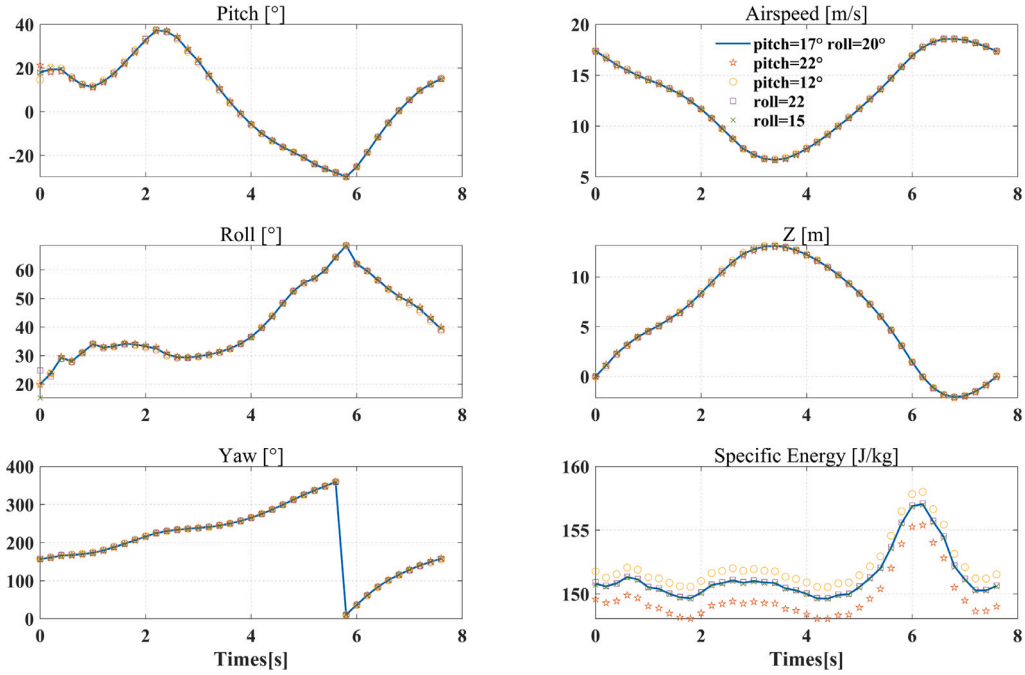
Recent studies have demonstrated the computational feasibility of using reinforcement learning for real-time UAV control. Chowdhury and Keshmuri (2024) equipped a fixed-wing UAV with an NVIDIA Jetson Nano, featuring a 1.42 GHz Quad-core CPU and 4 GB memory, which is capable of running a PPO-based RL algorithm efficiently. Similarly, multicopters have been successfully trained to perform tasks under disturbances using an RL framework. In real-world experiments, the onboard computer executes the RL algorithm while the attitude is controlled by a cascaded PID controller (Yu et al., 2023). This architecture, which combines RL with the lower-level controller, is conceptually aligned with the approach proposed in this paper. Another study (Wang et al., 2023) has shown that the computational load of SAC can be reduced by distributing tasks between local hardware and remote workstations. For example, action computation can be performed locally, while policy updates can be offloaded to a remote server. A similar distributed architecture could be adopted in our framework to balance latency and computational load, ensuring compatibility with onboard hardware constraints and real-time performance.

7. Conclusion

Dynamic soaring has showcased the promising vision of perpetual flight for unmanned aerial vehicles. However, in order to investigate the energy harvesting mechanisms, the dynamics models and controllers have often been simplified. This exclusion of the tracking controller in the process of designing optimizers has presented challenges when attempting to implement dynamic soaring scenarios in real-world applications. Therefore, this study proposes an iterative framework that utilizes HITL based on DRL to autonomously execute the Rayleigh cycle. We revisit previous methodology and expand upon prior work. To closely simulate real-world conditions, the flight dynamics, lower-level control system, wind field, and aerodynamic calculations were integrated into a high-fidelity environment. The HITL method combined with DRL in the iterative framework is utilized to solve the dynamic soaring problem. Experimental results indicate that the trained agent acts as a higher-level guider, steering the lower-level controller to execute the Rayleigh cycle without energy consumption, bringing the glider back close to its initial state. Our findings support the notion that the glider accumulates more energy in the presence of nonlinear wind characteristics compared to a linear wind field. A



(a) Deviations of 5% from standar the take-off position



(b) Deviations of 5% from standar the take-off attitude

Fig. 14. State of trajectories with intentionally set deviations before the start. The blue color represents the reference trajectory, while the points represent the others.

key novelty of this study is the simultaneous search for the optimal trajectory and control of the glider. We argue that the proposed framework is well-suited for deployment in real-world applications, which provides smooth action variations, accommodates small action limits, and integrates well with the lower-level controller.

Robustness tests are conducted as wind conditions are uncontrollable in real-world environments. Random wind fields are introduced to investigate the resilience of the agent against disturbances. Our analysis reveals that the entropy objective function in DRL exhibits a certain level of resilience to random variations in the wind field. Furthermore, tests are conducted to simulate deviations in the takeoff

state, representing drift and bias in the onboard sensors. Our findings confirm that deviations in attitude have minimal impact on the agent. Regarding the tests on initial position bias, it is observed that drift in the south and east direction leads to deviations from the initial position. Overall, results are promising and support the conclusion that agents complete the trajectory of the Rayleigh cycle. Considering the satisfactory performance of DRL algorithms even in weak wind environments, our future focus will be on the deployment of onboard computers on gliders. It can be a further effective scheme for applications of dynamic soaring in the real world.

The potential scope of the proposed framework extends beyond the scope considered in our work. The modular design of the framework allows for customization and modification, making it suitable for diverse aircrafts that implement DRL algorithms. Considering the expenditure of time associated with training, future research could prioritize optimizing the algorithms to ensure compatibility with various aircrafts, thereby improving the efficiency of algorithm deployment. To enhance adaptability and versatility in addressing the diverse demands of dynamic soaring in different scenarios, a comprehensive automatic system can be envisioned, incorporating multiple agents.

CRedit authorship contribution statement

Yufeng Lu: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fangfang Xie:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition. **Tingwei Ji:** Writing – review & editing, Validation, Supervision, Resources, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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